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Digital Twin Model Reconstruction and Environmental Load Analysis of the Umbrella-Shaped Tensile Membrane Structure Based on 3D Point Clouds

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Abstract

Actual construction errors and accumulated damage seriously affect the spatial structural form of the tensile membrane structure, making it difficult to characterize the full-life spatial form of the physical tensile membrane structure based on the original theoretical model. There are still significant challenges in achieving the virtual–real correspondence of digital twin results based on the original theoretical design model. It is necessary to propose a high-precision digital twin model construction method that adapts to the full life cycle of the tensile membrane structure. For this reason, a refined digital twin model reconstruction and morphological deviation visualization method for the umbrella-shaped tensile membrane structure combined with 3D point clouds, computational geometry algorithms, and 3D3S finite element simulation modeling is proposed in this paper. Firstly, three-dimensional point cloud data of the umbrella-shaped membrane structure test bench were acquired, and the point cloud data were accurately registered based on the Iterative Closest Point (ICP) algorithm. Secondly, the physical membrane surface reconstruction was obtained by applying the Screened Poisson Surface Reconstruction (SPSR) algorithm. Subsequently, the 3D3S theoretical model of the umbrella-shaped tensile membrane structure was updated according to the contour of the measured membrane surface reconstruction model. Finally, global spatial geometric morphological deviation was compared among the theoretical models and the reconstructed physical model. Furthermore, different environmental load combinations were analyzed to support hazard warning of the refined digital twin model of the physical umbrella-shaped tensile membrane structure. The results show that the maximum spatial form deviation of the original theoretical model was 43.56 mm, whereas the maximum form deviation of the updated theoretical model was only 0.05 mm. The updated theoretical model accurately reflected the spatial form characteristics of the physical membrane structure and satisfied the requirement for digital twin physical–virtual consistency. In addition, the global stress distribution between the original/updated theoretical models differed markedly, and the updated theoretical model is more suitable for identifying weak areas and evaluating safety performance of the actual umbrella-shaped tensile membrane structure under different ultimate environmental loads.

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Keywords: digital twin; 3D point cloud; three-dimensional model reconstruction; spatial geometric morphological deviation; environmental load analysis; tensile membrane structure

1. Introduction

A digital twin is a virtual counterpart of a physical entity that supports bidirectional and real-time data mapping and seamless switching between physical and virtual domains [1,2]. Grieves [1] first proposed a three-dimensional model integrating the physical entity, the virtual entity, and their interactive connection. With the development of application demands, digital twin models have been continuously updated for different application scenarios [2]. Tao [3] proposed a five-dimensional digital twin model and systematically analyzed its application domains, levels, disciplines, dimensions, generality, and functions. Owing to its integrated monitoring, control, and prediction capabilities, digital twin technology has become prominent in manufacturing [4]. Against the background of domestic intelligent development strategies, digital twin technology has gradually expanded from manufacturing to the construction industry and has become one of the key technologies supporting the transformation of traditional construction toward intelligent construction [5].

Although research on digital twins in the construction field has grown rapidly, existing studies remain highly fragmented and regionally oriented [6]. Congiu [7] developed a digital twin system framework to support facility managers in decision-making by effectively integrating Building Information Modeling (BIM) and Internet of Things (IoT). For intelligent construction services, Pan [5] developed a detailed digital twin framework containing physical models, virtual models, and connected data by integrating BIM, IoT, and data mining (DM) techniques. Oti [8] used laser scanners and cameras to acquire real-time building data and established a building energy-consumption feedback loop, which reduces the gaps between different building life-cycle stages through continuous correction of design and facility management in the digital twin environment. Yu [9] proposed an intelligent operation and maintenance scheme for urban roads based on multi-source spatiotemporal monitoring data. Song [10] developed a conceptual framework that integrates digital twins with BIM, aiming to extend BIM applications from design and construction stages to the entire life cycle of bridge engineering through real-time data interaction and intelligent asset management. Based on data integration, synchronization, and analysis, Lu [11] designed a building–city digital twin model to support anomaly detection, environmental monitoring, maintenance optimization, prioritization, and energy planning. Qi [12] proposed a digital twin framework for transmission towers by integrating BIM, IoT, and the Finite Element Method (FEM), which supports analysis of structural health monitoring and visual warning under wind loads and flood scouring. These studies indicate that the integration of digital twins with DM, IoT, BIM, and FEM can enable multi-source data fusion and in-depth data mining, and further reduce carbon footprints, improve indoor environmental quality, and enhance resource utilization and energy efficiency [13,14].

Although existing studies show that digital twins are evolving from static BIM toward dynamic decision-support systems, simulation modeling methods remain central to their implementation. Digital modeling technology is a core method for virtual-environment modeling and is also a primary approach for digital twin modeling. Based on geometric modeling technology, a virtual mapping platform for physical entities can be established and formed as data carriers by constructing geometric models of various physical entities. Digital twins currently have moved from theoretical research to engineering applications,

and their successful implementation depends on the construction of digital twin models adapted to specific objects [15]. Liu [16] proposed a digital twin modeling method based on biomimicry and realized adaptive construction of a multi-physics digital twin for machining processes. Jian [17] proposed a STEP-based 3D model recommendation method that integrates quaternion semantic diffusion with parameter diffusion to improve the efficient exchange, retrieval, and reuse of digital twin models. Arvanitis [18] proposed a robust and computationally efficient 3D saliency mapping method that integrates spectral and geometric information to accurately identify significant geometric features in noisy industrial 3D models. Achour [19] developed a collaborative semantic mapping framework that integrates asynchronous semantic maps generated by multiple mobile robots to incrementally update and synchronize digital twins of controlled indoor environments. Gopalakrishnan [20] introduced process modeling and characterization technology to capture stress distributions in manufacturing processes and applied it to turbine-disk component design and digital twin life-cycle analysis. Hu [21] constructed an integral blisk digital twin model, and proposed a real-time calibration method for blisk dynamic characteristics. Liang [22] proposed a real-time displacement perception method, and constructed a theoretical displacement perception model based on multi-point camera observations, which enables one to drive the twin model to map the real state of the physical environment and supports intelligent algorithms for manufacturing-state prediction and control.

Compared with conventional building structures, inspection and evaluation technologies for the tensile membrane structure remain relatively underdeveloped, making it difficult to provide long-term and effective technical support for safe operation. Analysis of the above studies indicates that the primary task in implementing a digital twin is to create an accurate reconstructed digital twin model for the application object. An accurate structural twin model has become an indispensable tool for assessing load-bearing capacity, safety, and maintenance during the structural full life cycle [23,24]. In order to conduct reliable load-effect analysis and structural safety assessment for tensile membrane structures, the key requirement is to establish an accurate reverse-modeled digital twin model that conforms to the spatial form of the real in-service structure. Up to now, a common problem in the design and construction of domestic tensile membrane structures is the disconnection between construction and structural design, which leads to serious accumulation of construction errors. In addition, environmental corrosion, material aging, load fatigue, and overload effects caused by emergencies may occur during service life, resulting in damage accumulation and resistance degradation and thus significantly affecting the structural spatial geometric form. Therefore, when structural performance analysis and safety inspection are carried out for a physical tensile membrane structure, the original theoretical design model can no longer adequately represent the physical structure. Digital simulation results based on the original theoretical design model may differ considerably from those of the actual structure, making it difficult to construct a high-fidelity model and achieve physical–virtual consistency. Therefore, it is necessary to use intelligent monitoring methods to perform refined reconstruction of the physical tensile membrane structure, and to support full-life-cycle morphology monitoring and safety inspection.

To address these challenges, this paper integrates three-dimensional point clouds, 3D3S modeling, geomagic modeling, and FEM environmental load analysis to investigate refined digital twin model reconstruction, spatial form deviation visualization, and nonlinear structural analysis under different load combinations for an umbrella-shaped tensile membrane structure. This work can provide a reference for the full-life-cycle monitoring of digital twins in tensile membrane structures. The main contributions are as follows:

1. A digital twin model test platform for the umbrella-shaped tensile membrane structure was built in this work, achieving real-time acquisition of three-dimensional spatial point cloud data information of the physical umbrella-shaped tensile membrane structure.
2. By integrating 3D point cloud scanning technology, ICP algorithm, SPSR algorithm, 3D3S modeling, and Geomagic modeling methods, a refined digital twin model and an updated structural theoretical model for the umbrella-shaped tensile membrane structure are constructed. This enables visualization of spatial geometric form deviations in the entire domain of the umbrella-shaped tensile membrane structure.
3. Based on the original/updated theoretical model of the umbrella-shaped tensile membrane structure, the changes in the bearing capacity of the umbrella-shaped tensile membrane structure theoretical models under different load-combination conditions are investigated, which provides the possibility for realizing danger warning of the physical umbrella-shaped tensile membrane structure.

2. Methodology

The tensile membrane structure has the typical characteristics of a large-scale and complex configuration. Traditional measurement methods struggle to meet the data requirements of full scale and high accuracy. 3D reverse modeling based on 3D point cloud can not only achieve fast and comprehensive spatial information collection, but also provide data support for structural construction quality control, deformation analysis, and structural health monitoring. It can efficiently generate digital twin reconstruction models that are consistent with physical entities, which provides a data foundation and model basis for performance evaluation and optimization design of existing structures.

2.1. 3D Point Cloud Registration Methodology

An independent three-dimensional coordinate system is established when an object is scanned using a laser scanning system. In this coordinate system, the laser scanner is generally taken as the coordinate origin o, the laser scanning direction is defined as the Y axis, the axis in the scanning plane perpendicular to the Y axis is defined as the X axis, and the Z axis is perpendicular to the scanning plane. α is the horizontal scanning angle, representing the rotation angle of the laser beam in the horizontal plane relative to a reference direction, and typically rotating about the vertical axis of the scanner. θ is the vertical scanning angle, representing the elevation or depression angle of the laser beam relative to the horizontal plane. The horizontal scanning angle and vertical scanning angle θ can then be obtained using a precision-clock-controlled encoder. Each coordinate point can be calculated as shown in Equation (1).

$$\begin{cases} X = R\cos\theta\cos\alpha \\ Y = R\cos\theta\sin\alpha \\ Z = R\sin\theta \end{cases} \quad (1)$$

where R is the slant distance between the point in the scanned object's point cloud and the coordinate origin O.

However, a single scanning station cannot capture structural full-scene three-dimensional point cloud information in most scenarios. Therefore, the laser scanner should be placed at multiple stations to collect structural global three-dimensional geometric information. In general, a denser three-dimensional point cloud acquired by the laser scanner provides a more complete representation of the scanned object. According to the laser scanning system principle described above, the point cloud data acquired from multiple scanner stations are located in different relative coordinate systems. Therefore, multiple groups of point cloud data should be registered to construct a complete and accurate point

cloud model of the scanned object. As a commonly used precise point cloud registration method, the Iterative Closest Point (ICP) algorithm was proposed by Besl in 1992 [25]. This method repeatedly searches for the nearest point correspondence between two sets of point clouds on the basis of existing 3D point cloud data, and estimates the optimal rigid body transformation, so that the source point cloud gradually approaches the target point cloud. As the mainstream rigid body registration algorithm for precise registration of flexible surface point clouds, the ICP algorithm is adapted to the rigid registration of the three-dimensional point cloud data of the umbrella-shaped tensile membrane structure.

Define the source point cloud (local point cloud of the membrane surface to be spliced) as $P = \{p_i \in \mathbb{R}^3,\ i = 1, 2, \ldots, n\}$ and the target point cloud (reference membrane surface point cloud) as $Q = \{q_i \in \mathbb{R}^3,\ j = 1, 2, \ldots, n\}$. The objective of the ICP algorithm is to solve for the optimal rotation matrix $R \in SO(3)$ and translation vector $t \in \mathbb{R}^3$, and minimize the mathematical expression of the mean square error of the Euclidean distance corresponding to the two sets of point clouds, as shown in Equation (2). The corresponding target loss function is constructed as shown in Equation (3).

$$E(R, t) = \sum_{i=1}^{N} \left\| Rp_i + t - q_j \right\|^2 \tag{2}$$

$$(R^*, t^*) = \arg\min_{R,t} \sum_{i=1}^{N} \left\| Rp_i + t - q_j \right\|^2 \tag{3}$$

where R is the rotation matrix. t is the translation vector. $Rp_i + t$ is the new coordinate obtained by transforming the source point p_i through the rotation matrix R and translation vector t. $Rp_i + t - q_j$ is the three-dimensional error vector between the transformed source point and its corresponding target point. N is the number of valid corresponding point pairs participating in the current iteration. R^* is the optimal rotation matrix, and t^* is the optimal translation vector.

The evaluation index based on three-dimensional point cloud rigid transformation estimation is mainly used to evaluate the accuracy between the point cloud obtained by rigid transformation and the true value. The commonly used evaluation index is root mean square error, which is mathematically expressed as Equation (4). The root mean square error represents the average absolute error between the true value and the predicted value.

$$\mathrm{RMSE} = \sqrt{\frac{1}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2} \tag{4}$$

where N is the number of corresponding point cloud data, y_i is the transformed point cloud data, and $\hat{y}_i$ is the source point cloud data.

2.2. Three-Dimensional Membrane Reconstruction Theory

Membrane surface reconstruction of the umbrella-shaped tensile membrane structure is based on the three-dimensional point cloud geometric topology information of the physical membrane surface to reconstruct the spatial geometric form characteristics. Poisson surface reconstruction algorithms are widely valued for high-fidelity scene reconstruction capabilities [26]. The main idea of the Poisson surface reconstruction algorithm is to establish an intrinsic relationship between the scanned object point cloud sample set S and the scalar indicator function χ of the scanned object, so that the surface reconstruction can be transformed into a benign sparse dynamic Poisson equation solving problem, and then the reconstructed target object can be obtained by extracting the isosurface. Briefly, Poisson surface reconstruction algorithms transform the reconstruction into a global problem, and complete the spatial reconstruction of the directed point set by solving the Poisson function

and extracting isosurfaces. When reconstructing the surface of the target object based on the Poisson surface reconstruction algorithm, an intrinsic relationship is established between the sample model point cloud set S and the implicit function χ to establish an isosurface threshold γ, as shown in Equation (5). The energy functional mathematical expression corresponding to the Poisson surface reconstruction algorithm is shown in Equation (6).

$$S = \left\{ x \in \mathbb{R}^3 \,|\, \chi(x) = \gamma \right\} \tag{5}$$

$$E(\chi) = \int_{\Omega} \|\nabla\chi(x) - V(x)\|^2 dx \tag{6}$$

The Poisson surface reconstruction method was first proposed by Kazhdan in 2006 [26]. Afterwards, this team further improved the Poisson surface reconstruction algorithm and proposed the screened Poisson surface reconstruction (SPSR) algorithm [27]. Compared with the performance of the traditional Poisson surface reconstruction (TPSR) algorithm, the SPSR algorithm not only improved the processing speed, but also reduced the algorithm computational time complexity. The SPSR algorithm, as a subsequent improved version of the TPSR algorithm, explicitly adds point position constraints outside the Poisson smoothing term to improve the geometric fidelity of the reconstructed model and reduce the possibility of the reconstructed model deviating from the original sampling points due to excessive smoothing. Its corresponding energy function expression is shown in Equation (7). Therefore, the SPSR algorithm is adopted as the main algorithm for the three-dimensional reconstruction of the membrane surface in the umbrella-shaped tensile membrane structure in this paper, and the TPSR algorithm is applied as a comparison case.

$$E(\chi) = \int_{\Omega} \|\nabla\chi(x) - V(x)\|^2 dx + \lambda \sum_{i=1}^{N} w_i (\chi(p_i) - c_i)^2 \tag{7}$$

where $\int_{\Omega} \|\nabla\chi(x) - V(x)\|^2 dx$ is the smoothness term of the surface reconstruction algorithm, which is used to ensure the overall smoothness and normal consistency of the reconstructed curve. $\lambda \sum_{i=1}^{N} w_i (\chi(p_i) - c_i)^2$ is a positional constraint term used to constrain the implicit function to satisfy a given function value near the sampling point. λ is the filtering weight that controls the balance between fitting point clouds and smoothing curves. w_i is the point weight, and c_i is the implicit function value of the target corresponding to the sampling point.

In terms of 3D reconstruction evaluation metrics, the point cloud coverage ratio, radius variation coefficient, and height standard deviation are commonly used as performance indicators to evaluate surface reconstruction algorithms. The point cloud coverage ratio represents whether the points in the input point cloud are covered by the reconstructed mesh surface. The coefficient of variation of the radius indicates that there is almost no obvious ellipticization, local convexity or boundary distortion at the edge of the reconstructed structure. The height standard deviation implies that the reconstructed structure has a smooth and stable edge, and the reconstruction algorithm does not produce significant fluctuations in Equation (9) and Equation (10), respectively.

$$C_k = \frac{1}{N} \sum_{i=1}^{N} [d(p_i, S_R) < k\bar{s}] \times 100\% \tag{8}$$

$$CV_\tau = \frac{\sigma_\tau}{\bar{\tau}} \times 100\% \tag{9}$$

$$\sigma_z = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(z_i - \bar{z})^2} \tag{10}$$

where N is the number of points participating in the evaluation in the input point cloud; p_i is the i-th point in the input point clouds; S_R is the reconstructed mesh surface; $d(p_i, S_R)$ is the shortest Euclidean distance from the point to the reconstructed surface S_R; $\bar{s}$ is the average point spacing or average sampling spacing of the input point clouds; k is the dimensionless distance threshold coefficient; and $k\bar{s}$ is the distance threshold to judge whether the point cloud is effectively covered by the reconstructed surface. $\bar{r}$ is the average of the radius of all edge sampling points, and σ_τ is the standard deviation of the radius of edge sampling points. z_i is the height of the i-th sampling point relative to the reference plane, and $\bar{z}$ is the average height of all participating evaluation points.

2.3. Finite Element Theoretical Model Modification Theory

Finite element model modification is usually achieved by optimizing model parameters to ensure that the predicted finite element results are consistent with the measured target results within the allowable deviation range. The model modification in classical structural dynamics is often based on modal test results such as frequency and mode shape. However, the large-span tensile membrane structure adopts coated fabric membrane materials and auxiliary support structures to form a stress-controlled spatial structure through the tensioning method, as shown in Equation (11). From the perspective of structural mechanics, the core difference between tensile membrane structures and traditional rigid structures lies in their highly nonlinear mechanical characteristics of shape changing with stress. The spatial geometric configuration state of the membrane surface is an equilibrium form constrained by both prestress and boundary conditions. Therefore, the measured three-dimensional point cloud of the membrane surface is not only geometric data, but also indirectly reflects the comprehensive influence of various factors such as construction errors, material degradation, and prestress loss.

As shown in Equation (12), the membrane surface reconstructed based on three-dimensional point clouds described in Section 2.2 does not actually have mechanical properties. Therefore, it is necessary to use 3D3S simulation software, Version 10 [28] combined with a membrane surface reconstruction model based on the 3D point cloud, in order to minimize the shape error between the theoretical model and the measured physical model, as shown in Equation (13). In response to the finite element theoretical model modification of the umbrella-shaped tensile membrane structure studied in this paper, the membrane surface coordinates reconstructed from the 3D point cloud were selected as the target data to modify the finite element model. The updated theoretical model is close to the spatial geometry of the membrane surface and satisfies the mechanical equilibrium of the tensile membrane structure. Compared to the original theoretical model, the stress distribution of the updated theoretical model is closer to the actual membrane surface of the umbrella-shaped tensile membrane structure.

$$\mathbf{S}_0 = \mathbf{S}(\mathbf{u}_0; \boldsymbol{\delta}_0) \tag{11}$$

$$\mathbf{S}_R = \left\{ x \in \mathbb{R}^3 \,\middle|\, f_\theta(x) = 0 \right\} \tag{12}$$

$$\delta^* = \arg\min_{\delta} \mathbf{D}[\mathbf{S}_0(\delta),\, \mathbf{S}_R(x)] \tag{13}$$

where $\mathbf{S}_0$ is the initial theoretical membrane surface, which is the theoretical design membrane surface before correction; $\mathbf{u}_0$ is the initial displacement state vector; $\boldsymbol{\delta}_0$ is the initial model parameter vector; $\mathbf{S}_R$ is the reconstructed membrane surface based on three-

dimensional point clouds; x represents the position vector in three-dimensional space; $f_\theta(x)$ is the continuous implicit scalar function reconstructed from the screened Poisson surface; θ is the parameter set of the implicit function f_θ. δ^* is the modified model parameter vector; $D[]$ is the distance measure between the theoretical membrane surface model and the three-dimensional reconstructed membrane surface.

2.4. Global Deviation Analysis Between the Theoretical Reference Model and the Physical Measurement Model

Geomagic software, version 2014 has been widely used in research fields such as 3D metrology, detection, and deviation quantification due to its powerful 3D computing capabilities. Geomagic software also developed software interfaces for CAD and CAM, which can achieve data sharing and fast processing between multiple software through setting plugins, and complete 2D or 3D geometric measurement graphic comparison. When conducting global deviation analysis on two models in Geomagic software, the essence is to analyze their geometric distance fields in three-dimensional space. When comparing the two models in three dimensions, geomagic software calculates a deviation value for each data point in the measurement data; that is, each measurement point has a measurement position P_m. By projecting the direction, the reference position P_r is found, and then the gap vector is obtained by pointing P_r to P_m. The deviation value of this point is then calculated, and the mathematical expressions are shown in Equations (14) and (15). In this work, Geomagic software is applied to quantitatively analyze and compare the spatial geometric deviations among the original theoretical model, the physically measured reconstruction model and the updated theoretical model of the umbrella-shaped tensile membrane structure.

$$\vec{g}_i = P_{m,i} - P_{r,i} \tag{14}$$

$$d_i = \|\vec{g}_i\| \tag{15}$$

where $P_{m,i}$ is the i-th sampling point on the measurement model; $P_{r,i}$ is the corresponding position point on the reference model; $\vec{g}_i$ is the gap vector; and d_i is the deviation value of the comparison sampling point.

2.5. The Overall Technical Flowchart of the Proposed Method

Construction errors and accumulated damage significantly affect the spatial geometric configuration of the tensile membrane structures. Consequently, digital simulation results based on the original theoretical design model are difficult to use for constructing a high-fidelity model, and physical–virtual consistency of the structural digital twin cannot be achieved. In this paper, an umbrella-shaped tensile membrane structure was used as the research object, and a digital twin test platform was established in this work. The refined membrane surface digital twin model reconstruction and hazard warning were investigated by integrating three-dimensional point cloud data, the ICP algorithm, the SPSR algorithm, Geomagic software, and 3D3S finite element nonlinear load analysis. The overall technical route of this paper is shown in Figure 1.

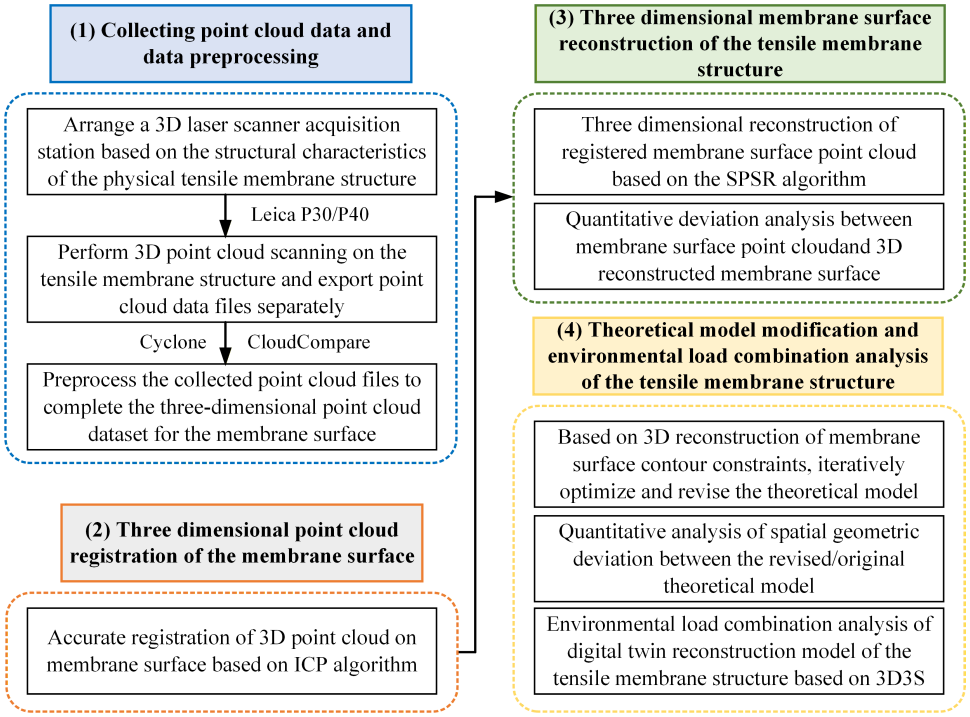


Figure 1. Overall technical flowchart of the method proposed in this paper.

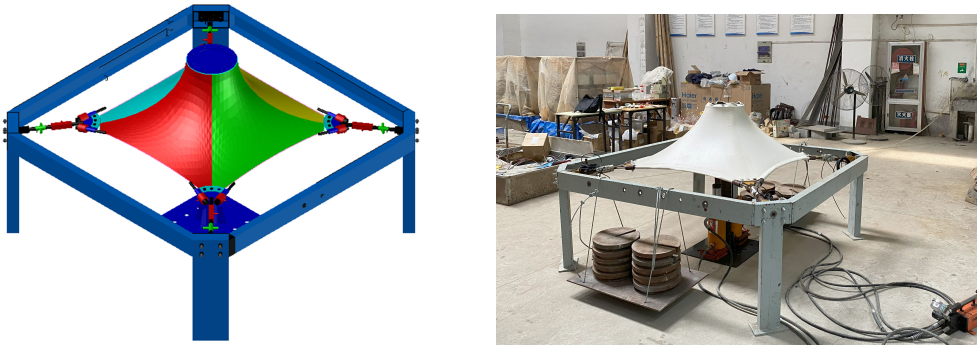
3. Data Collection and Preprocessing

3.1. Digital Twin Test Bench for the Umbrella-Shaped Tensile Membrane Structure and 3D Point Cloud Data Acquisition

A Leica P30/P40 ultra-high-speed three-dimensional laser scanner (Leica Geosystems AG, Heerbrugg, Switzerland) was used to perform three-dimensional point cloud scanning of the umbrella-shaped tensile membrane structure. The technical parameters of the Leica P30/P40 ultra-high-speed three-dimensional laser scanner are shown in Table 1. In this paper, the rise span ratio of the umbrella-shaped tensile membrane structure built independently is 0.4, the top cap size is 300 mm, the longitudinal height is 380 mm, and the bottom side length of the square membrane surface is 950 mm. The original theoretical model and the built physical structure test bench of the umbrella-shaped tensile membrane structure are shown in Figure 2. The membrane material used in the umbrella-shaped tensile membrane structure test bench is 250T PVC membrane material, and the mechanical parameters of the membrane material are shown in Table 2.

Table 1. Technical parameters of the Leica ultra-high-speed 3D laser scanner.

Equipment Model	Leica P30/P40 Ultra-High-Speed Three-Dimensional Laser Scanner	Effective Scanning Distance	270 m@34% Reflectivity
Angle measuring accuracy	8″	Laser category	Class I
Distance measuring accuracy	1.2 mm + 10 ppm	Target acquisition accuracy	2 mm@50 m
Scanning speed	1,000,000 points/s	Airborne target acquisition range	≥75 m
Noise accuracy	0.5 mm@50 m	Precision biaxial compensation accuracy	1.5″



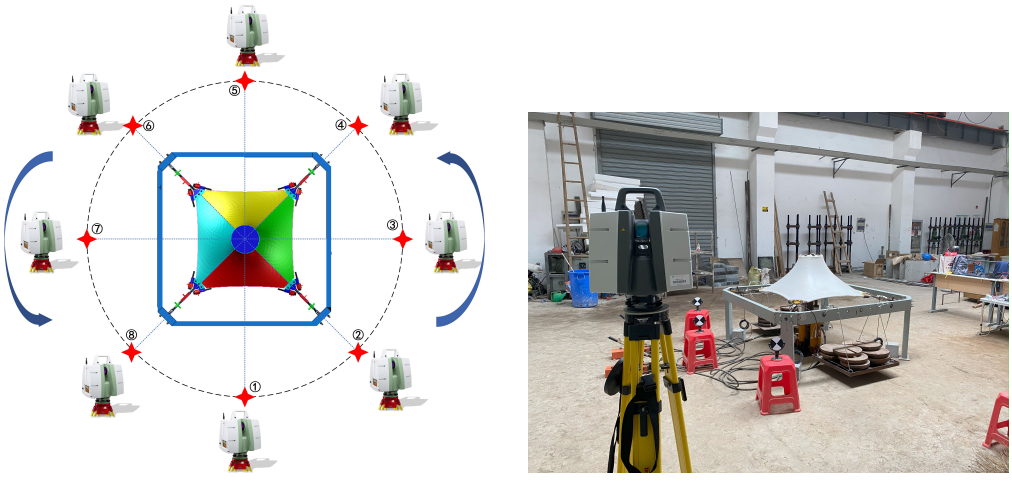
(**a**) (**b**)

Figure 2. Schematic diagrams of the original theoretical model and the physical structure test bench of the umbrella-shaped tensile membrane structure. (**a**) The original theoretical design model test bench. (**b**) The built physical structure test bench.

Table 2. Mechanical parameters of the 250T PVC membrane material.

Material Category	Thickness (mm)	Areal Density (g/m^2)	Tensile Strength (kN/m)		Elastic Modulus (N/mm)		Poisson's Ratio
			Warp Direction	Weft Direction	Warp Direction	Weft Direction	
250T	0.87	1046	112	94	996	840	0.2

In terms of structural configuration, the umbrella-shaped tensile membrane structure was a special-shaped curved structure with an irregular spatial form. Owing to the perspective limitation, its complete structural information could not be obtained from a single scan during three-dimensional point cloud acquisition. Therefore, different stations were arranged in a ring around the tensile membrane structure, and the structure was scanned in batches at specified angular intervals. Considering the structural scale of the experimental platform established in this paper, each acquisition station was arranged at every 45-degree interval around the umbrella-shaped tensile membrane structure, as shown in Figure 3.



(**a**) (**b**)

Figure 3. Station design and field test for acquiring the three-dimensional point cloud of the umbrella-shaped tensile membrane structure. (**a**) Design scheme for eight scanning stations. (**b**) Field site of 3D point cloud scanning.

To ensure that the acquired point cloud data of the tensile membrane spatial structure had the same shape and size and differed only in coordinate position, the scanning equipment was kept horizontally fixed throughout the acquisition process. This prevented

shaking or movement and avoided non-rigid transformations among the scanned point cloud datasets, which could otherwise affect subsequent registration. The point cloud data of the umbrella-shaped tensile membrane structure acquired from eight different stations using the three-dimensional laser scanner were processed using Cyclone point cloud analysis software, Version 9.2.0 provided with the Leica scanner,(Leica Geosystems AG, Heerbrugg, Switzerland) as shown in Figure 4.

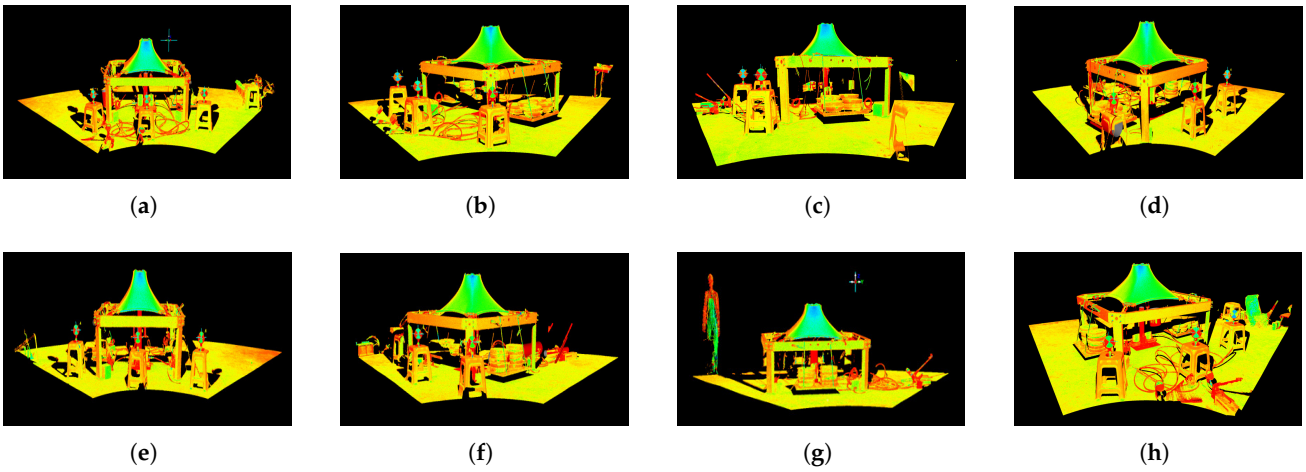


Figure 4. Three-dimensional point cloud data of the umbrella-shaped tensile membrane structure from eight different stations. (**a**) Station 1. (**b**) Station 2. (**c**) Station 3. (**d**) Station 4. (**e**) Station 5. (**f**) Station 6. (**g**) Station 7. (**h**) Station 8.

3.2. Preprocessing of 3D Point Cloud Data for the Umbrella-Shaped Tensile Membrane Structure

As shown in Figure 5, the initial three-dimensional point cloud data acquired from the eight stations effectively characterizes the spatial geometric form of the physical umbrella-shaped tensile membrane structure. However, a large amount of noise data was inevitably acquired during point cloud acquisition, including non-target point clouds from objects other than the observed umbrella-shaped tensile membrane structure, such as other laboratory equipment, and moving objects. Noise points among the initial point cloud data were manually removed by using Cyclone software, and the membrane surface point cloud data of eight stations after data preprocessing are shown in Figure 5.

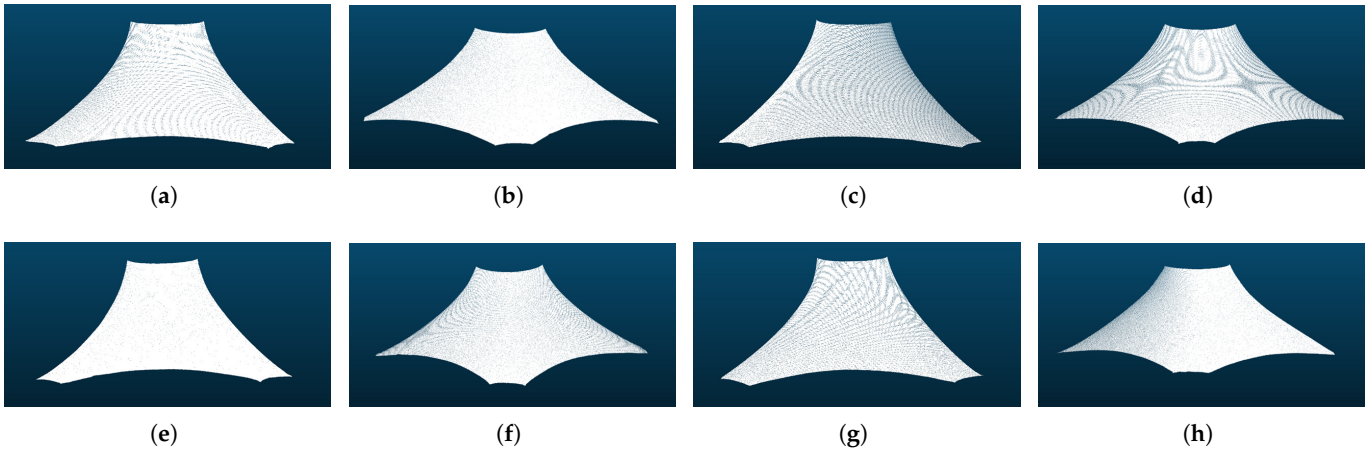


Figure 5. Three-dimensional point cloud data of the umbrella-shaped tensile membrane structure from eight stations after data preprocessing. (**a**) Station 1. (**b**) Station 2. (**c**) Station 3. (**d**) Station 4. (**e**) Station 5. (**f**) Station 6. (**g**) Station 7. (**h**) Station 8.

4. Experiment and Discussion

4.1. 3D Point Cloud Data Registration of the Umbrella-Shaped Tensile Membrane Structure

After the above point cloud data were obtained, point cloud registration was required to construct a complete three-dimensional point cloud model of the umbrella-shaped tensile membrane structure. In this paper, a point cloud registration method based on FPFH feature coarse registration, point-to-surface ICP registration, and pose map global optimization is adopted to achieve the stitching of multi-view membrane surface point clouds. The specific parameters and process of registering membrane surface point clouds using the ICP algorithm based on point-to-surface reconstruction are shown in Figure 6. Point cloud registrations were carried out based on the membrane surface point cloud dataset of the umbrella-shaped tensile membrane structure, and the visualization and registration indicator results are shown in Figure 7 and Table 3.

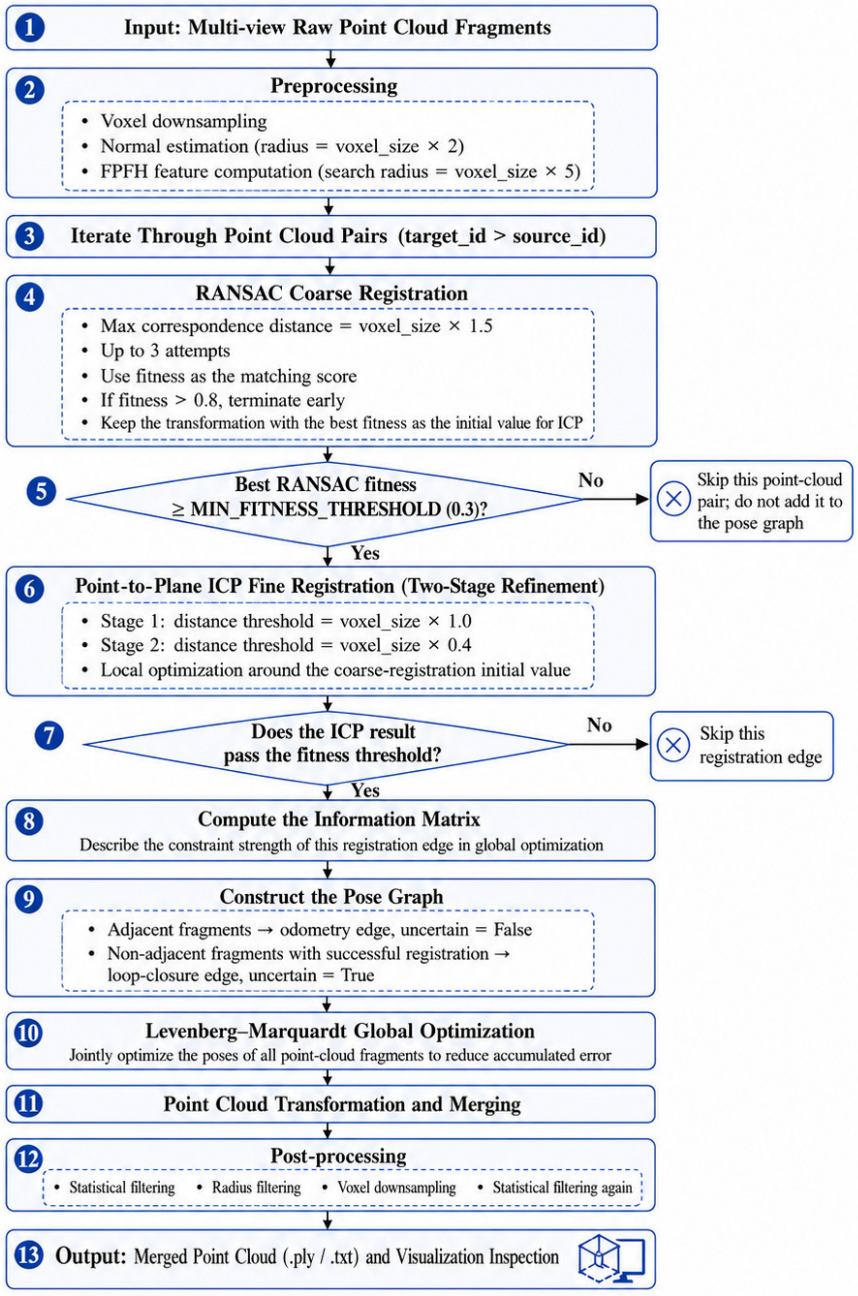
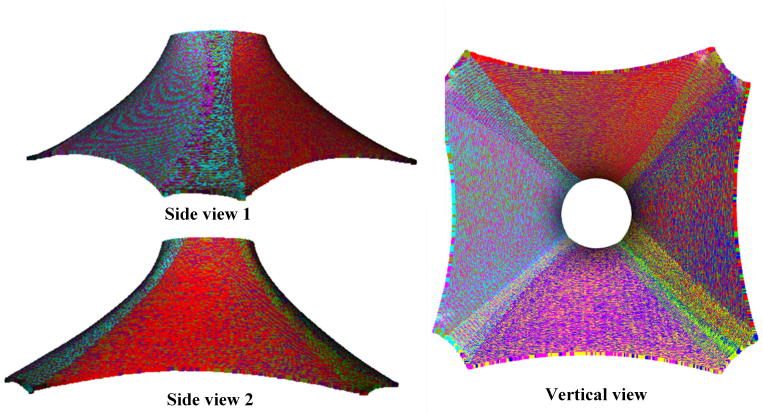


Figure 6. Parameters and flowchart of the ICP algorithm adopted in this paper.

(**a**)



(**b**)

Figure 7. Registration effect of the membrane surface point cloud of the umbrella-shaped tensile membrane structure. (**a**) Initial positions of the membrane surface point clouds at different sites before registration. (**b**) Spatial geometric shape of the membrane surface point cloud after registration.

Table 3. Comparison of registration indicators for membrane surface point clouds at eight stations based on ICP algorithm.

Station 1 & Station 2		**Station 1 & Station 3**		**Station 1 & Station 4**		**Station 1 & Station 5**	
Fitness	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**
0.6005	0.002050	0.8733	0.004422	0.6381	0.003400	**0.9576**	**0.002800**
0.5900	0.001847	0.5379	0.002550	0.5142	0.002364	**0.8717**	**0.002208**
Station 1 & Station 6		**Station 1 & Station 7**		**Station 1 & Station 8**		**Station 2 & Station 3**	
Fitness	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**
0.5964	0.002905	0.9436	0.003567	0.5875	0.002977	**0.9032**	**0.002019**
0.5396	0.002178	0.7204	0.002411	0.5122	0.002167	**0.8881**	**0.001831**
Station 2 & Station 4		**Station 2 & Station 5**		**Station 2 & Station 6**		**Station 2 & Station 7**	
Fitness	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**
0.9343	0.003471	0.8767	0.002790	**0.9890**	**0.002865**	0.8876	0.002959
0.7492	0.002398	0.7860	0.002118	**0.8796**	**0.002286**	0.7845	0.002198
Station 2 & Station 8		**Station 3 & Station 4**		**Station 3 & Station 5**		**Station 3 & Station 6**	
Fitness	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**
0.9783	**0.003072**	0.6076	0.002017	0.8895	0.004692	0.5697	0.002570
0.8451	**0.002278**	0.5989	0.001843	0.4907	0.002508	0.5313	0.002218
Station 3 & Station 7		**Station 3 & Station 8**		**Station 4 & Station 5**		**Station 4 & Station 6**	
Fitness	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**
0.9676	**0.003241**	0.5627	0.003092	0.8294	0.002037	0.8937	0.003672
0.7763	**0.002482**	0.4847	0.002274	0.8145	0.001821	0.7049	0.002451
Station 4 & Station 7		**Station 4 & Station 8**		**Station 5 & Station 6**		**Station 5 & Station 7**	
Fitness	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**
0.9531	**0.002582**	**0.9334**	**0.002588**	0.5527	0.002013	0.9316	0.003992
0.9039	**0.002353**	**0.8783**	**0.002278**	0.5446	0.001832	0.6176	0.002520

Table 3. *Cont.*

Station 5 & Station 8		Station 6 & Station 7		Station 6 & Station 8		Station 7 & Station 8	
Fitness	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**	**Fitness**	**RMSE**
0.5716	0.002719	0.8883	0.002083	**0.9676**	**0.003184**	0.5683	0.002016
0.5285	0.002311	0.8677	0.001820	**0.8246**	**0.002287**	0.5595	0.001831

As shown in Figure 7 and Table 3, the point cloud data of each station has achieved good coincidence in the overall spatial position when the registration of membrane point cloud data obtained from different measurement stations is completed based on the ICP algorithm. The point cloud distribution in the main area of the membrane surface is continuous, and the outer boundary and central opening contour are clear. The spatial curvature and overall undulating shape of the membrane surface are well preserved in the lateral view. This also indicates that the ICP algorithm can effectively eliminate the initial position and pose differences between the clouds of multiple measurement stations, and the registered point cloud basically reflects the real spatial shape of the umbrella-shaped tensile membrane structure, which can provide a reliable data basis for the membrane surface reconstruction and morphological deviation analysis.

It can also be found from Figure 7 that a small number of discrete points and reduced local coincidence can still be observed near the local edge, the central opening and the area with sparse point clouds of the membrane surface. The main reason for this phenomenon is the lack of complete neighborhood geometric constraints in the edge and opening area of the membrane surface, and the limited visible range and effective overlapping area of different stations at these positions. This makes it difficult for the ICP algorithm to establish a stable and unique point correspondence. Because the ICP algorithm relies on the nearest neighbor correspondence and aims to minimize the overall distance error, local low overlap, non-uniform density and abnormal points may make some point clouds converge to the local optimal position and present local discrete phenomena.

4.2. Membrane Surface Reconstruction in the Umbrella-Shaped Tensile Membrane Structure

After obtaining the complete point cloud data of the registered membrane surface, the three-dimensional model of the membrane surface is required to be reconstructed based on the spatial geometric features. The SPSR algorithm is adopted as the main algorithm for the three-dimensional reconstruction of the membrane surface in the umbrella-shaped tensile membrane structure in this paper. The corresponding parameter settings and membrane surface three-dimensional reconstruction process are shown in Figure 8. The TPSR algorithm and the SPSR algorithm were adopted to perform membrane surface reconstruction of the umbrella-shaped tensile membrane structure. The corresponding reconstruction results are shown in Figures 9 and 10 and Table 4.

As illustrated in Figure 9, the TPSR algorithm can effectively reconstruct the spatial geometry of the membrane surface during 3D reconstruction, and the overall membrane surface reconstruction is relatively smooth and stable. However, the TPSR algorithm has certain shortcomings in expressing membrane surface boundaries and local details. Due to the emphasis on global smoothness in this method, it is easy to weaken the local geometric changes in the original membrane surface point cloud during the reconstruction process, which results in excessive smoothing of the membrane surface edge region, and incomplete preservation of some boundary undulations and local curvature features. It is worth noting that both boundary morphology and local curvature changes are important features for the umbrella-shaped tensile membrane structure, which reflect the spatial morphology of the membrane surface. Therefore, the TPSR algorithm has certain limitations in high-precision representation of the spatial geometric form of the membrane surface.

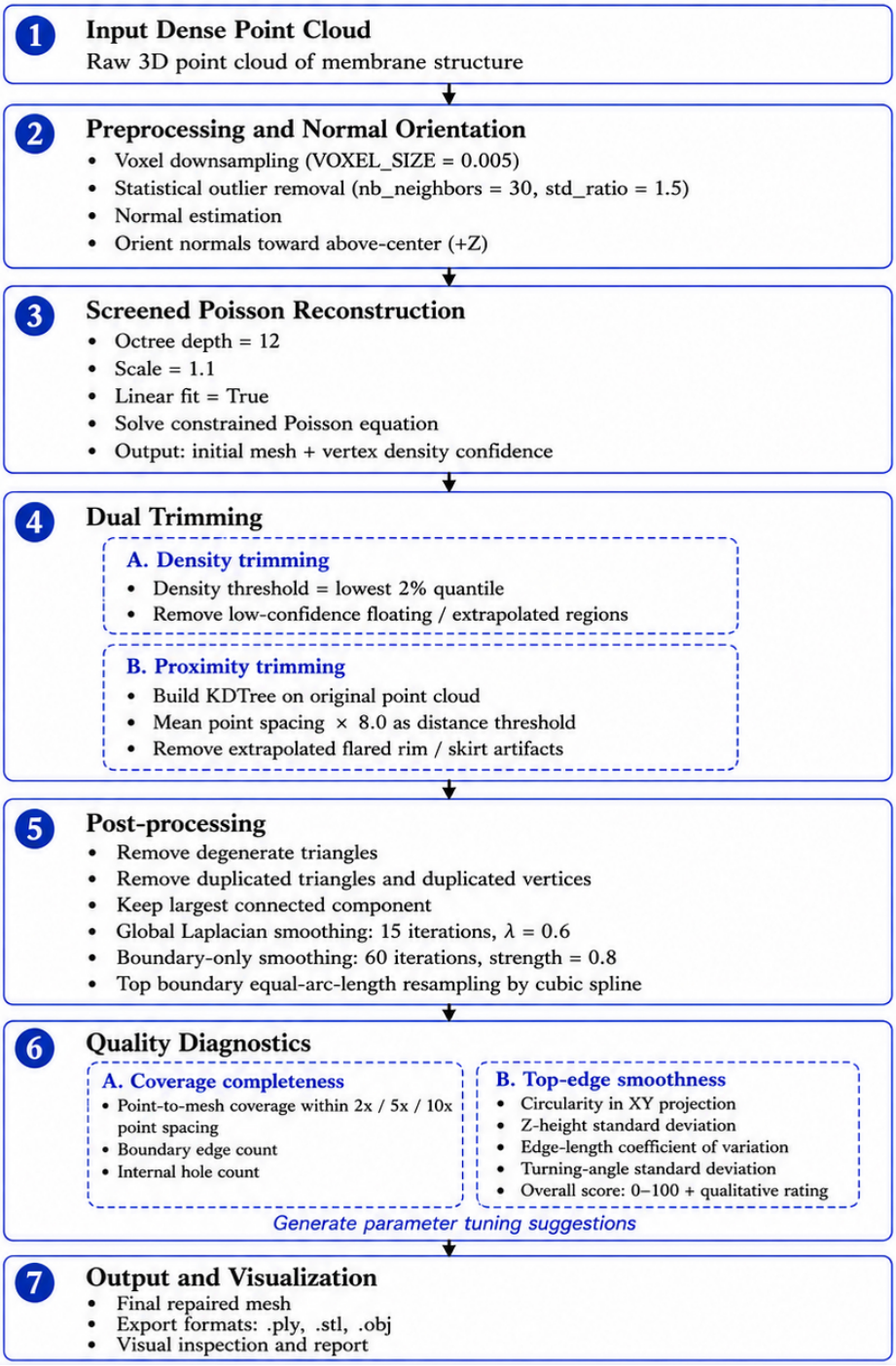
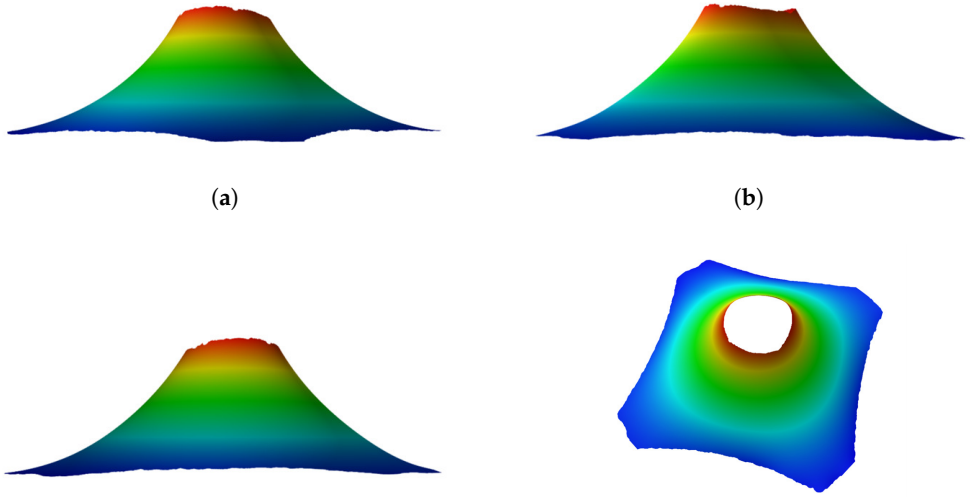


Figure 8. Schematic diagram of the calculation process for the SPSR algorithm.

Compared with the TPSR algorithm, the SPSR algorithm further introduces point cloud position constraints during the reconstruction process, making the reconstructed surface closer to the original sampling points while satisfying normal field consistency. As shown in Figure 10, the SPSR algorithm has a better reconstruction effect on the membrane surface spatial geometric form of the umbrella-shaped tensile membrane structure. The geometric features of the membrane surface bulge area, curvature transition area, and boundary droop area are more obvious, and the overall membrane surface reconstruction has a higher degree of compatibility with the original point cloud data. This indicates that the SPSR algorithm has better performance in preserving membrane surface details and restoring local morphology. Therefore, the SPSR algorithm can more effectively restore the spatial geometric features of the membrane surface while maintaining the overall continuity.

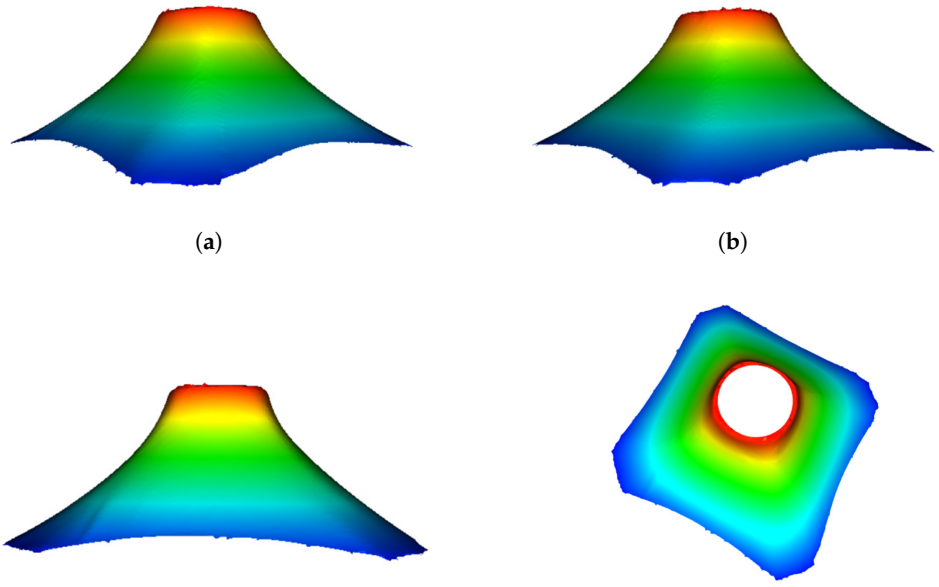
The SPSR algorithm performs better in terms of original point cloud compatibility, local detail preservation, and boundary morphology expression, and is highly suitable for high-precision 3D reconstruction tasks of the membrane surface in the umbrella-shaped tensile membrane structure.



(**a**) (**b**)

(**c**) (**d**)

Figure 9. Membrane surface reconstruction effect based on the TPSR algorithm. (**a**) Side view 1. (**b**) Side view 2. (**c**) Side view 3. (**d**) Side view 4.



(**a**) (**b**)

(**c**) (**d**)

Figure 10. Membrane surface reconstruction effect based on the SPSR algorithm. (**a**) Side view 1. (**b**) Side view 2. (**c**) Side view 3. (**d**) Side view 4.

In order to comparatively study the influence of different parameters in the SPSR algorithm on the membrane surface reconstruction quality, this paper adjusts the model depth to control the value in Equation (7), thereby governing the overall smoothness of the reconstructed membrane surface. Additionally, the point cloud sampling voxel size is tuned to reflect the value in Equation (7), which controls the overall fitting accuracy of the reconstructed membrane surface. The calculation results are presented in Table 5.

Table 4. Surface reconstruction indicator comparison for membrane surface point clouds based on the TPSR algorithm and the SPSR algorithm.

Algorithm	C_k	CV_τ	σ_ε	Computation Time
TPSR algorithm	55.10% ± 0.14%	6.88% ± 0.10%	0.00698 ± 0.0008	32.5 s ± 0.24 s
SPSR algorithm	100% ± 0.00%	1.13% ± 0.03%	0.00448 ± 0.00006	16.23 s ± 0.10 s

Table 5. Parameter sensitivity analysis of membrane surface reconstruction performance based on the SPSR algorithm.

Depth	Voxel Size	C_k	CV_r	σ_z	Computation Time
10	0.005	100%	1.14%	0.00440	8.44 s
11	0.005	100%	1.13%	0.00439	10.39 s
12	0.005	100%	1.15%	0.00448	16.23 s
13	0.005	100%	1.13%	0.00447	25.97 s
14	0.005	100%	1.12%	0.00440	36.36 s
0.002	12	100%	1.94%	0.00473	28.56 s
0.003	12	100%	2.10%	0.00625	35.71 s
0.004	12	100%	2.26%	0.00446	29.21 s
0.005	12	100%	1.13%	0.00448	16.23 s
0.006	12	100%	2.26%	0.00446	25.32 s

As can be seen from Table 5, when the voxel size is fixed at 0.005, as the depth increases from 10 to 14, the point cloud coverage ratio remains 100% throughout, the radius variation coefficient fluctuates only within 1.12% to 1.15%, and the height standard deviation is also stable in the range of 0.00439–0.00448, indicating that the reconstruction accuracy is not sensitive to changes in the depth parameter. However, the train time increases significantly from 8.44 s to 36.36 s, suggesting that the computational efficiency is more sensitive to the depth parameter. In contrast, when the depth is fixed at 12, varying the voxel size from 0.002 to 0.006 also does not affect the reconstruction completeness, but the radius variation coefficient exhibits obvious fluctuations between 1.13% and 2.26%, and the height standard deviation increases to a maximum of 0.00625, indicating that the voxel size has a more pronounced effect on the reconstruction accuracy and stability of the membrane surface. Among the tested voxel sizes, a value of 0.005 yields a relatively low radius variation coefficient and height standard deviation, with a computation time of only 16.23 s. Therefore, considering reconstruction completeness, geometric accuracy, and computational efficiency comprehensively, a depth of 12 and a voxel size of 0.005 constitute the optimal parameter combination.

Further comparison was made between the reconstructed membrane surface based on two surface reconstruction algorithms and the measured membrane surface 3D point cloud. The deviation distributions between the reconstructed membrane surface and the measured membrane surface point cloud are shown in Figure 11.

From Figure 11a,c, it can be observed that the TPSR algorithm can effectively reconstruct the membrane surface of the umbrella-shaped tensile membrane structure with good continuity and smoothness. However, the overlap comparison with the original membrane surface point cloud shows that the TPSR algorithm has significant geometric deviations in the edge and top opening areas of the membrane surface, and local areas exhibit surface expansion and excessive smoothness. The reason for this phenomenon is that the TPSR algorithm mainly relies on global implicit functions for solving, emphasizing the overall continuity of the membrane surface, and has relatively insufficient constraint capabilities for open boundaries and local high-curvature areas. Therefore, although the TPSR algorithm can obtain a relatively complete three-dimensional surface model in the

task of reconstructing the membrane surface of the umbrella-shaped tensile membrane structure, its ability to preserve the local geometric features of the original point cloud is limited, and there is still significant room for improvement in reconstruction accuracy and boundary fidelity.

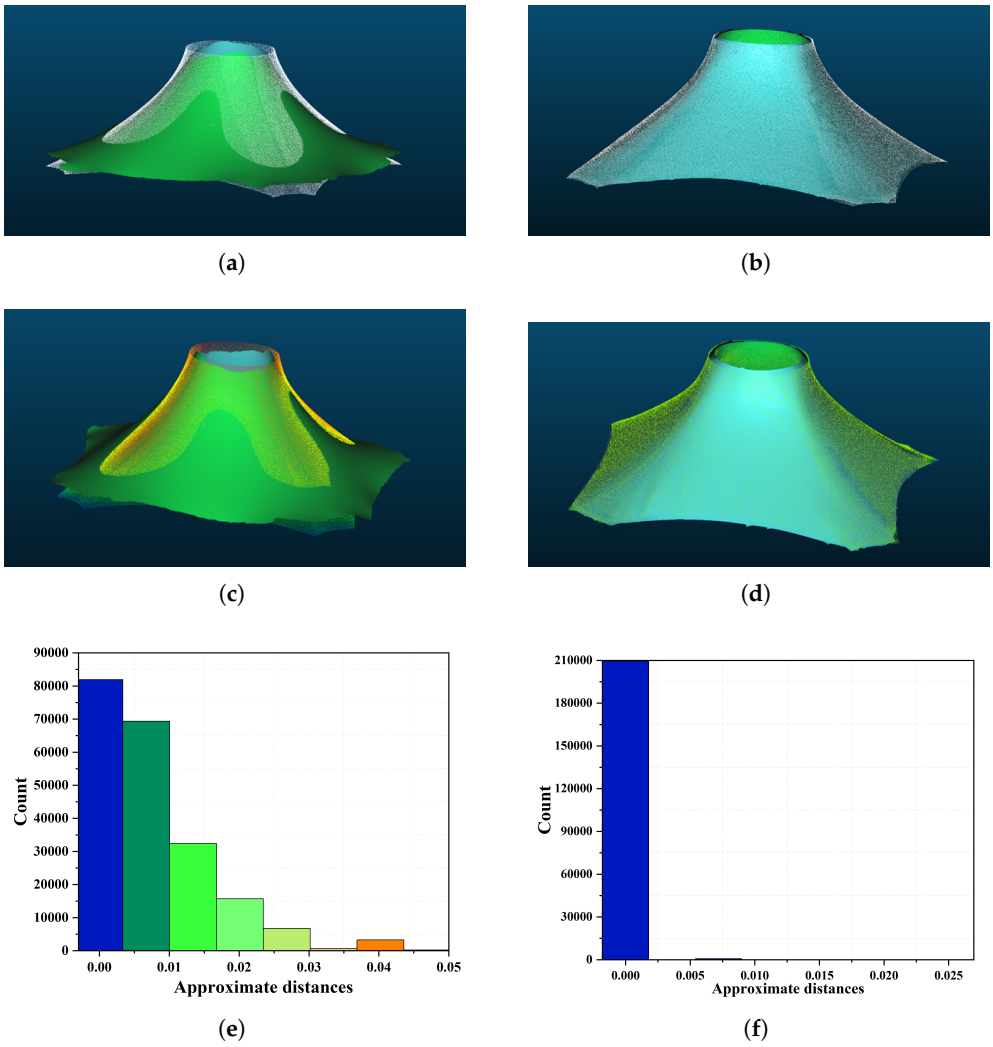


Figure 11. Spatial geometry form comparison between 3D reconstructed membrane surface model and measured registered membrane surface point cloud. (**a**) Comparison between membrane surface reconstruction based on the TPSR algorithm and complete point cloud of the measured membrane surface. (**b**) Comparison between membrane surface reconstruction based on the SPSR algorithm and complete point cloud of the measured membrane surface. (**c**) Distance deviation between the reconstructed membrane surface based on the TPSR algorithm and measured membrane surface complete point cloud. (**d**) Distance deviation between the reconstructed membrane surface based on the SPSR algorithm and measured membrane surface complete point cloud. (**e**) Statistical histogram of distance deviation between the reconstructed membrane surface based on the TPSR algorithm and measured membrane surface complete point cloud. (**f**) Statistical histogram of distance deviation between the reconstructed membrane surface based on the SPSR algorithm and measured membrane surface complete point cloud.

From Figure 11b,d, it can be observed that the membrane surface model reconstructed based on the SPSR algorithm has a high degree of overlap with the original membrane surface point cloud. The overall surface shape of the reconstructed membrane surface is highly consistent with the actual spatial geometric features of the membrane surface of the umbrella-shaped tensile membrane structure. Compared to the TPSR algorithm,

the SPSR algorithm further enhances the constraints on the top and bottom boundary positions of the original point cloud while maintaining surface continuity and smoothness, enabling the reconstructed membrane surface to more accurately fit the input point cloud data. Specifically, in the main area of the membrane surface, near the top opening, and in areas with significant curvature changes, the SPSR algorithm exhibits better geometric preservation ability and local fitting accuracy. Therefore, the SPSR algorithm has higher reconstruction accuracy and better morphological fidelity in the task of reconstructing the membrane surface of the umbrella-shaped tensile membrane structure.

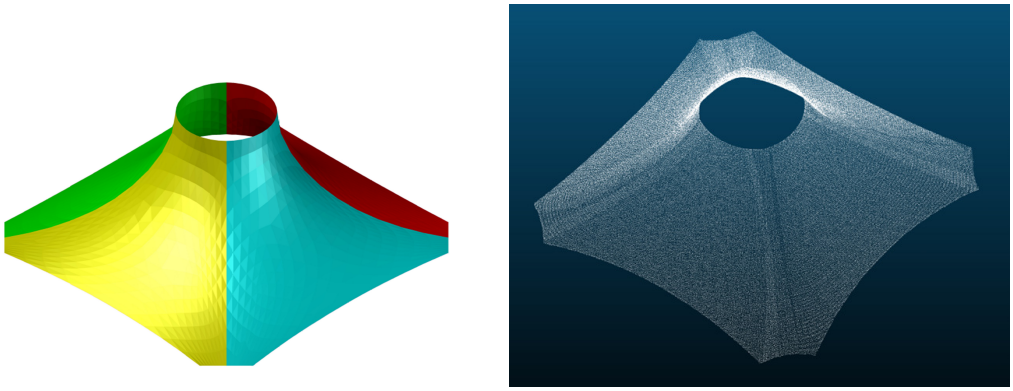
Comparisons among the original membrane point cloud data with a quantity of 210,273 and the reconstructed membrane surface based on two surface reconstruction algorithms are completed in this paper. It can be observed from Figure 11e that the reconstructed membrane surface deviation based on the TPSR algorithm is mainly concentrated in the low-deviation range, with approximately 81,918 point cloud deviations distributed in the low-value range close to 0. At the same time, there are still approximately 10,943 point clouds distributed in the high-deviation range, with a maximum deviation close to 0.05, which indicates that there is a significant geometric deviation in the local area of the model. As shown in Figure 11f, it can be seen that when comparing the original membrane surface point cloud with a quantity of 210,273 and the reconstructed membrane surface based on the SPSR algorithm, approximately 209,468 point cloud deviations are concentrated in the low-value range close to 0, which implies a high degree of consistency between the two models in terms of overall spatial position. Compared with the results of the TPSR algorithm, the deviation distribution based on the SPSR algorithm is more concentrated, the degree of dispersion is lower, and the number of high-deviation points is significantly reduced, which indicates that the registration accuracy between models is higher and the overall geometric deviation is smaller. The very few high-deviation points mainly come from local noise, edge points, or non-overlapping areas. Therefore, the overall consistency of the reconstructed membrane surface based on the SPSR algorithm is superior to the TPSR algorithm based on the deviation distribution characteristics.

4.3. Finite Element Theoretical Model Modification for the Umbrella-Shaped Tensile Membrane Structure

3D3S is a piece of domestic engineering software dedicated to the analysis and design of spatial structures [28,29]. It offers capabilities for parametric modeling, automatic form-finding, and nonlinear finite element analysis of cable–membrane structures. The software can account for the tensile properties of membrane materials, the cable–membrane coupling effect, and structural large-deformation behavior. It is an ideal analytical tool for simulating the mechanical response of tensile membrane structures under prestress and external loads. In this study, a computational model of the umbrella-shaped tensile membrane structure is established using 3D3S software. Then, form-finding analysis and load-response analysis are carried out to verify the geometric rationality of the updated theoretical model and the mechanical performance of the umbrella-shaped tensile membrane structure. Three-dimensional point cloud scanning technology can effectively complete the three-dimensional form reconstruction of the umbrella-shaped tensile membrane structure. The theoretical model, the physical structure, and the three-dimensional point cloud model of the umbrella-shaped tensile membrane structure investigated in this paper are shown in Figure 12.

As shown in Figure 12, it is worth noting that obvious deviations existed between the measured three-dimensional reconstructed model and the theoretical model of the umbrella-shaped tensile membrane structure. At this stage, only relying on the theoretical model could not accurately characterize the spatial form of the physical tensile membrane structure, nor could it be used to construct its digital twin model. Consequently, the spatial

tensile-force characteristics of the tensile membrane structure itself would be difficult to analyze. Therefore, the theoretical model of the umbrella-shaped tensile membrane structure was updated in this paper. Based on the external contour features of the measured three-dimensional point cloud reconstructed model and the membrane material parameters listed in Table 2, membrane form-finding was performed again on the basis of the original 3D3S theoretical form-finding framework. During the re-form-finding process, the external contour features of the measured three-dimensional point cloud reconstructed model were used as reference lines, so that the form-found theoretical membrane surface approached the external contour reference lines of the measured membrane surface. The full-field membrane stress distribution of the umbrella-shaped tensile membrane structure theoretical model after 3D3S form-finding was compared with that of the original theoretical model, as shown in Figure 13.



(**a**) (**b**)

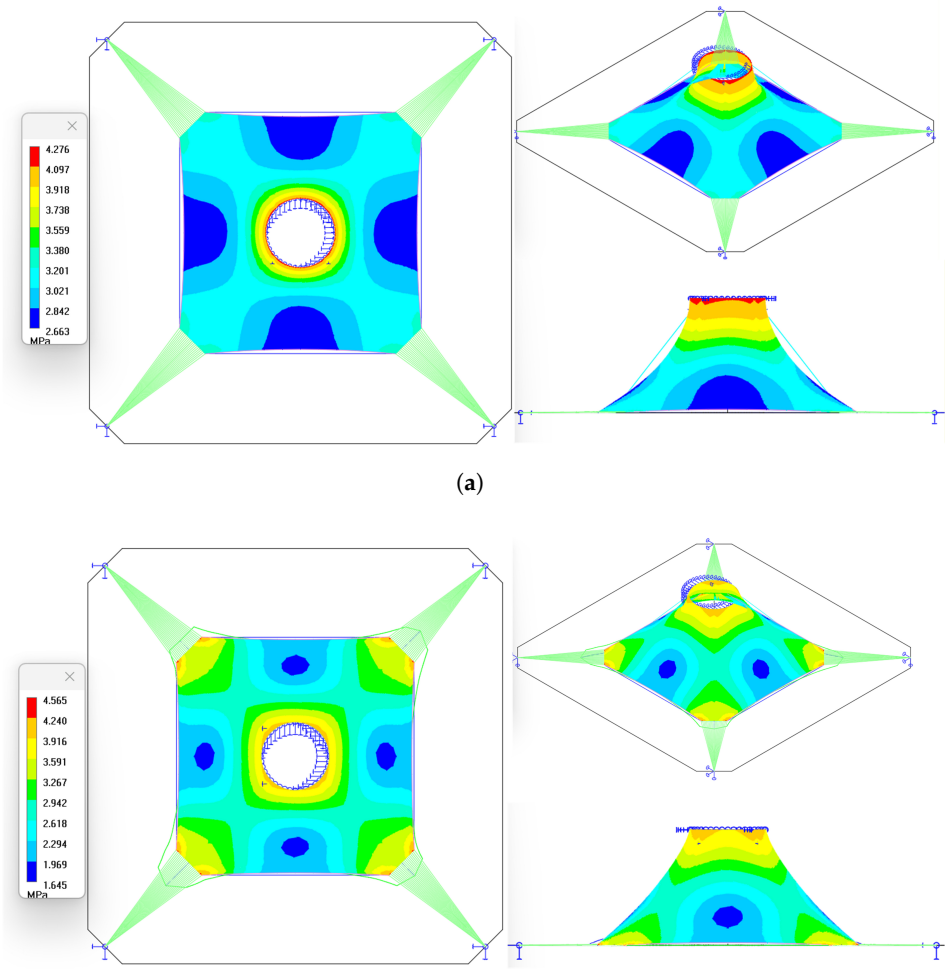
Figure 12. Comprehensive comparison of the original theoretical model and complete point cloud model of the membrane surface. (**a**) The original membrane surface theoretical model. (**b**) 3D point cloud of membrane surface after registration.

As shown in Figure 13, the average membrane stress distribution of the umbrella-shaped tensile membrane structure original theoretical model is approximately 2.8–3.3 MPa. The maximum membrane stress was approximately 4.3 MPa, and the maximum stress was distributed at the top of the umbrella-shaped tensile membrane structure. The overall membrane stress distribution was relatively uniform. After updating of the theoretical model, the average membrane stress distribution of the umbrella-shaped tensile membrane structure was approximately 2.3–3.5 MPa, the maximum membrane stress was approximately 4.6 MPa, and the maximum stress was distributed at the four corner points of the structure. Overall, a relatively large difference between the original/updated theoretical models is observed in the global membrane stress distribution.

Furthermore, comparison between the original/updated theoretical models of the membrane surface in the umbrella-shaped tensile membrane structure is illustrated in Figure 14.

It can be seen from Figure 14 that the updated theoretical model is basically the same as the original theoretical design model in terms of the overall three-dimensional configuration and boundary form, but the local geometry has changed significantly. Compared with the original theoretical design model, the transition from the central high point of the updated theoretical model to the surrounding membrane surface is more gentle. At the same time, the fluctuation and sagging degree in the middle and near the boundary of the membrane surface also changed to some extent, and the overall membrane surface curve was more continuous and coordinated. These differences imply that the measured membrane surface shape deviates from the ideal design model due to the influence of

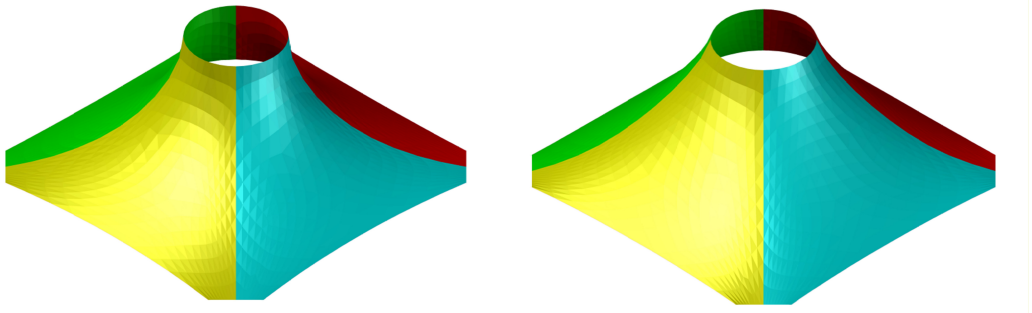
membrane material flexibility, pretension application error, support position deviation and construction forming conditions during the actual construction process. Relatively speaking, the updated theoretical model based on the reconstructed contour can more accurately reflect the morphological characteristics of the actual umbrella-shaped tensile membrane structure.



(**a**)

(**b**)

Figure 13. Stress distribution comparison between the original/updated theoretical models of the umbrella-shaped tensile membrane structure. (**a**) The original theoretical model. (**b**) The updated theoretical model.



(**a**) (**b**)

Figure 14. Comparison between the original/updated theoretical models of the umbrella-shaped tensile membrane structure. (**a**) The original theoretical model. (**b**) The updated theoretical model.

4.4. Global Form Deviation Analysis Between the Theoretical Design Model and the Actual Physical Model

To investigate spatial geometric form deviation between the theoretical/physical models, Geomagic software is used for verification. Firstly, the .stl file of the reconstructed membrane surface based on the SPSR algorithm was imported into Geomagic Studio. Subsequently, the encapsulated .stl file of the umbrella-shaped tensile membrane structure reconstruction model and the corresponding .stp file of the theoretical model were imported into Geomagic Control and placed in a unified coordinate system. Finally, the encapsulated .stl reconstruction model file of the umbrella-shaped tensile membrane structure was set as "TEST", and the .stp theoretical model file was set as "REF". The best-fit alignment and 3D comparison functions in Geomagic Control were then used to complete the morphological deviation analysis among the physical tensile membrane structure and the original/updated theoretical models. The comparison results are shown in Figures 15 and 16.

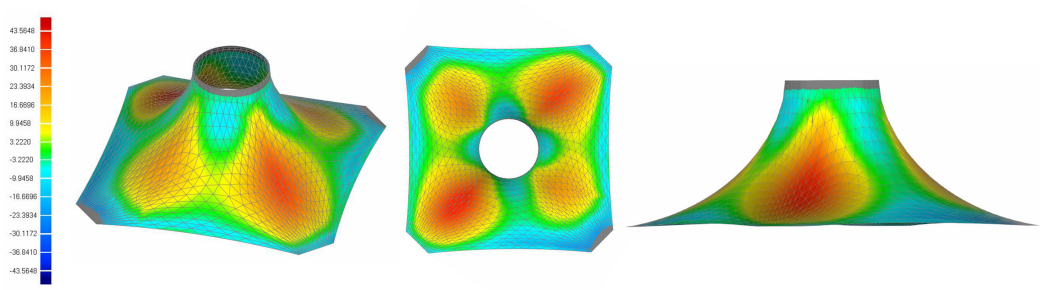


Figure 15. Morphological deviation comparison between the measured three-dimensional point cloud reconstructed model and the original theoretical model of the membrane surface.

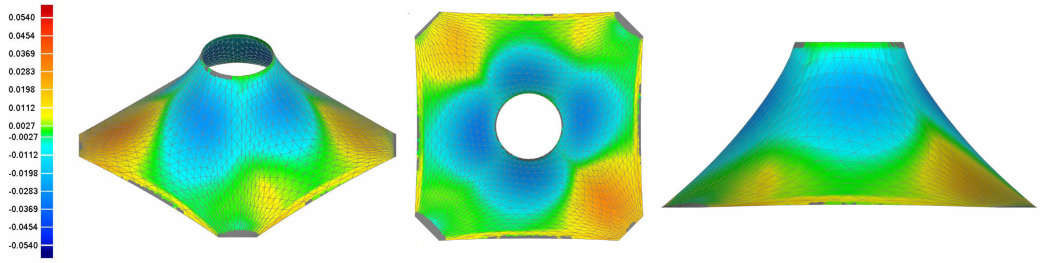


Figure 16. Morphological deviation comparison between the measured three-dimensional point cloud reconstructed model and the updated theoretical model of the membrane surface.

As shown in Figure 15, obvious deviations existed between the point cloud model and the original theoretical model of the umbrella-shaped tensile membrane structure in the middle and lower parts, and the deviations exhibited a paired distribution. The maximum deviation is 43.56 mm. According to the deviation color distribution map, the maximum deviation is located along one diagonal side of the umbrella-shaped tensile membrane structure. Analysis of the causes of the large deviation indicates that, during manual installation of the structure, the four gusset plates were installed along opposite diagonals. Because the corresponding threads of the gusset plates could not be guaranteed to be tightened to exactly the same fixed length, certain errors occurred when the four corner positions were fixed. In addition, unavoidable errors in membrane surface fabrication and the fact that the central support rod was not placed exactly at the center of the square frame because of uneven site conditions during platform construction were also major factors causing the form deviation between the physical membrane surface reconstruction model and the original theoretical membrane surface model.

As shown in Figure 16, the morphological deviation between the updated theoretical model and the measured three-dimensional point cloud reconstructed model changed markedly, and the maximum form deviation was only 0.05 mm. Except for a small form deviation along one diagonal side of the membrane surface, no obvious deviation is observed elsewhere, and the overall deviation distribution range is significantly reduced. This indicates that the updated theoretical model of the umbrella-shaped tensile membrane structure is in good agreement with the measured three-dimensional point cloud reconstructed model. In other words, the updated theoretical membrane surface model could correctly reflect the physical membrane structure and satisfy the requirement for physical–virtual consistency in the structural digital twin.

4.5. Ultimate Load Analysis of the Umbrella-Shaped Tensile Membrane Structure

Ultimate environmental load analysis was performed based on the original/updated umbrella-shaped tensile membrane structure models, respectively. Dead load, wind load, and snow load were simultaneously applied to the membrane surface to simulate the influence of adverse hazard conditions in the actual service environment on the full-field membrane stress. To obtain the full-field stress variation of the membrane surface under different load conditions, different partial factor parameters were set for the loads applied to the membrane structure model. Load cases including dead load, wind load, and snow load were arranged and combined in sequence to obtain several simulated hazard load combinations, as listed in Table 6.

Table 6. Simulated load combinations for the umbrella-shaped tensile membrane structure.

Scheme	Combination Under the Ultimate Limit State	Scheme	Combination Under the Serviceability Limit State
1	1.2 prestress + 1.2 dead load + 1.5 wind pressure	5	1.0 prestress + 1.0 dead load + 1.0 wind pressure
2	1.2 prestress + 1.2 dead load + 1.5 wind suction	6	1.0 prestress + 1.0 dead load + 1.0 wind suction
3	1.2 prestress + 1.2 dead load + 1.5 snow load	7	1.0 prestress + 1.0 dead load + 1.0 snow load
4	1.2 prestress +1.2 dead load	8	1.0 prestress +1.0 dead load

To realize intelligent construction of tensile membrane structures based on digital twins, the theoretical analysis model should be matched with the physical three-dimensional monitoring and reconstruction model. When the theoretical model is consistent with the actual structure, digital twin modeling of the membrane structure can be completed. Therefore, nonlinear analysis under different load combinations is required to compare the internal forces in the ultimate limit state and the stress distributions in the serviceability limit state of the original/updated theoretical models under each combination, as shown in Figures 17 and 18.

As shown in Figure 17, it can be seen from the stress cloud map under ultimate limit state environmental loads that there are significant differences in the stress distribution characteristics of the umbrella tensile membrane structure based on the original/updated theoretical model. The stress distribution on the membrane surface of the original theoretical model is generally regular, smooth, and symmetrical, showing strong idealized stress characteristics. The high-stress area is mainly concentrated near the central opening and the transition zone of tension at the corners. The stress distribution of the updated theoretical model is more complex, with more obvious differentiation of local high- and low-stress zones, and more prominent stress concentration phenomena around the central opening, boundary cable–membrane connection zone, and gusset plates. Under different ultimate

limit state load combinations, the maximum stress of the updated theoretical model does not uniformly increase monotonically. Among them, the maximum stress under load condition case 1 has increased, indicating that the modified model is more sensitive to local stress amplification effects. The maximum stress under load case 2 has decreased, but the stress is redistributed from the middle of the membrane surface to the boundary and corner regions. The maximum stress changes are relatively small under load case 3 and load case 4, but there are significant changes in the stress gradient and local stress distribution on the membrane surface. Overall, the model modification mainly changes the force transmission path and local stress distribution characteristics of the structure under ultimate load, making the calculation results more fully reflect the influence of actual geometric shape, boundary constraints, and cable–membrane synergy on the structural response. Therefore, the updated theoretical model is more suitable for identifying weak areas and evaluating safety performance of the umbrella-shaped tensile membrane structure under ultimate limit state environmental loads.

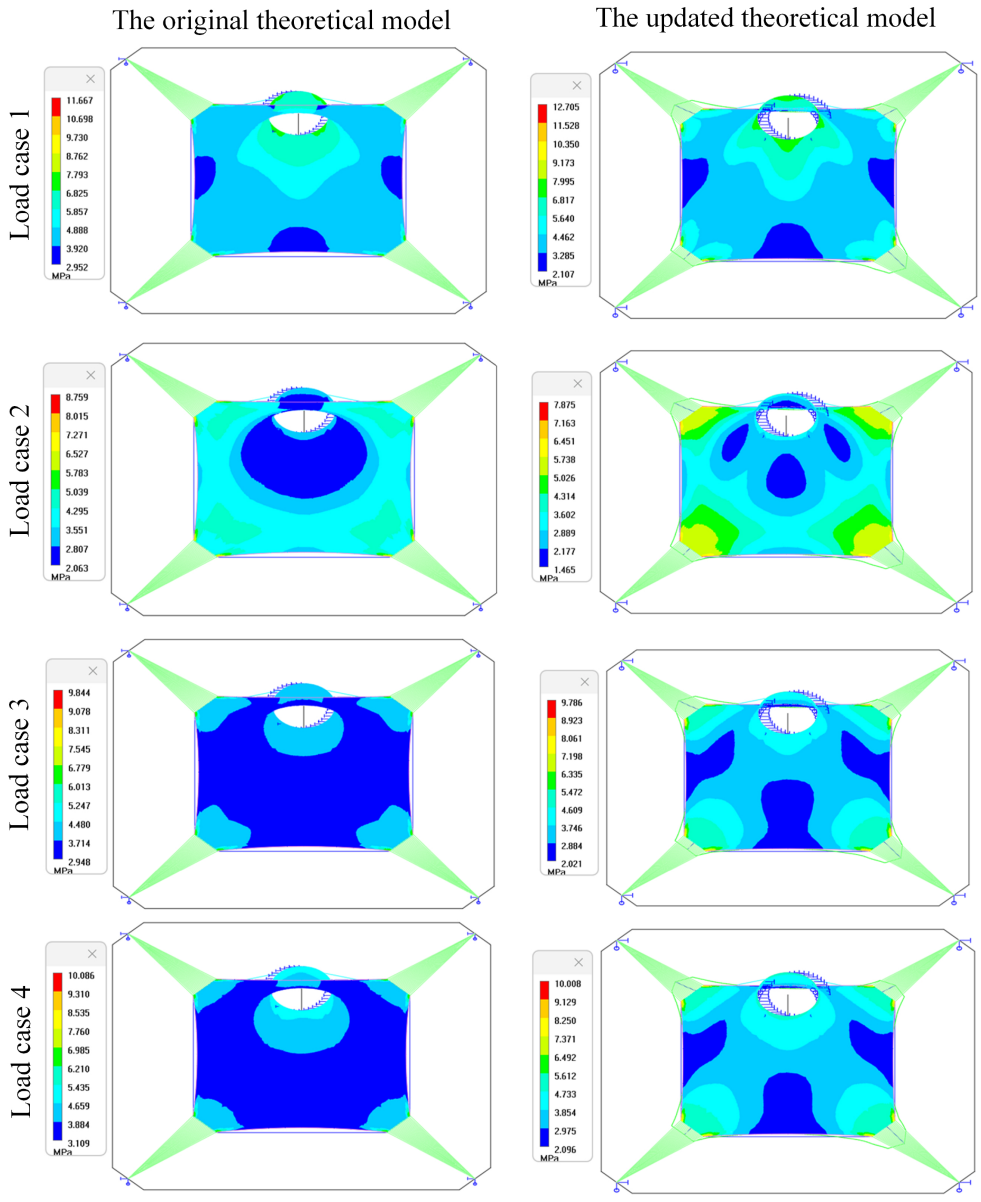


Figure 17. Stress distribution comparison of the umbrella-shaped tensile membrane structure under ultimate limit states for different load combinations.

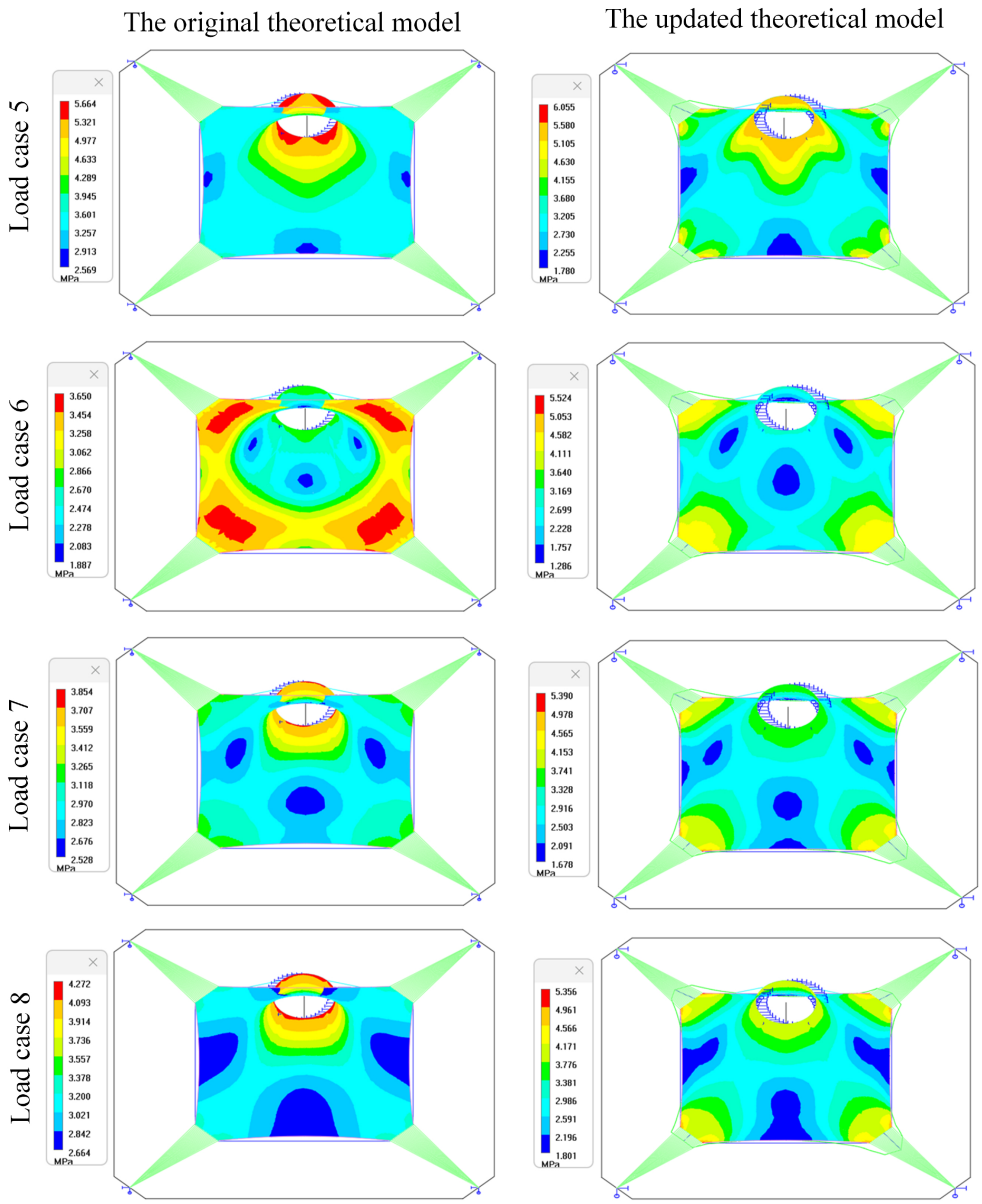
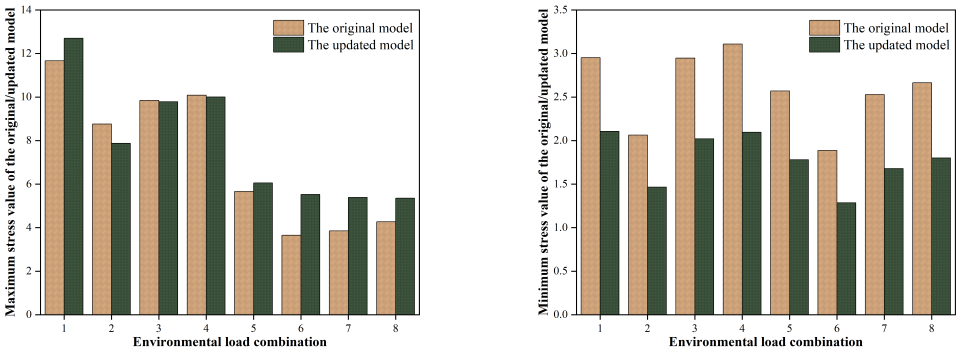


Figure 18. Stress distribution comparison of the umbrella-shaped tensile membrane structure under serviceability limit states for different load combinations.

As illustrated in Figure 18, it can be seen from the stress cloud map under serviceability limit state environmental load that there are significant differences in the stress distribution and peak response of the umbrella tensile membrane structure based on the original/updated theoretical model. The stress distribution on the membrane surface of the original theoretical model is generally regular and smooth, with high-stress areas mostly concentrated near the central opening and local corner areas, exhibiting strong idealized stress characteristics. The maximum stress of the updated theoretical model has increased under various load combinations, and the stress distribution is more discrete. The local stress concentration phenomena around the central opening, the boundary cable–membrane connection area, and the four-corner support area are more prominent. Among them, under load case 5, the high-stress zone of the modified model is still mainly located near the opening, but the boundary area is more significantly involved in the stress; the maximum stress increase is most significant under load case 6, indicating that model modification has a significant impact on the redistribution of internal forces under this condition; and under load case 7 and case 8, the updated theoretical model also exhibits

more pronounced local high- and low-stress zoning and stress gradient changes. Overall, the stress state of the structure under serviceability limit state environmental loads after model modification is more complex, which can more fully reflect the influence of actual geometric shape, boundary constraints, and cable–membrane synergy on the stress distribution on the membrane surface. Therefore, the updated theoretical model is more suitable for local stress evaluation and weak point identification during the normal service phase of the umbrella-shaped tensile membrane structure. The analysis results of the various combinations of loads mentioned above also indicate that, during the actual tensioning and forming process of the umbrella-shaped membrane structure, improper installation of the gusset plates may lead to potential stress concentration. It also demonstrates the necessity of constructing a digital twin theoretical simulation model for the umbrella-shaped tensile membrane structure.

The overall variation trends of the extreme principal stress of the membrane surface based on the original/updated theoretical models of the umbrella-shaped tensile membrane structure under different load combinations are compared, as shown in Figure 19.



(**a**) (**b**)

Figure 19. Comparison of stress values before and after updating of the umbrella-shaped tensile membrane structure under different load combinations. (**a**) Maximum membrane stress. (**b**) Minimum membrane stress.

As shown in Figure 19, under different load combinations, the maximum principal stress of the membrane surface in the updated theoretical model first decreased and then increased compared with that of the original model as the load cases changed in complexity. The corresponding minimum principal stress of the membrane surface remained consistently lower than the minimum principal stress of the original theoretical model. This indicates that a considerable performance difference existed between the physical umbrella-shaped tensile membrane structure and the theoretical design model under different load conditions. The maximum stress difference relative to the initial design stress is 2 MPa. The overall membrane stress distribution also differed considerably, which should be noted by researchers and engineers in the field of tensile membrane structures. To ensure that the theoretical model can be safely applied in practice, verification was required. By considering the internal forces of the membrane surface under the ultimate limit state and the serviceability limit state for various load combinations, the calculation results show that the original/updated theoretical models of the umbrella-shaped tensile membrane structure both satisfy the force requirements under different load combinations. This indicates that updating of the theoretical model is successfully completed and that digitization of the physical model is achieved. Therefore, this work provides support for intelligent construction and hazard-warning-related work for tensile membrane structures based on the digital twin platform.

5. Conclusions

Construction error accumulation during actual construction and continuous external environmental loads during service affect the spatial geometric form of tensile membrane structures. As a result, the original theoretical design model of a tensile membrane structure can no longer accurately reflect the real spatial form, and physical structural performance analysis and safety inspection cannot be effectively performed. To address this problem, an umbrella-shaped tensile membrane structure was used as the research case in this paper, and the main conclusions are as follows:

1. A digital twin model reconstruction test platform for the umbrella-shaped tensile membrane structure was independently established, and refined digital twin model reconstruction and spatial form deviation visualization were investigated based on three-dimensional point cloud scanning technology.
2. 3D point cloud scanning technology, ICP algorithm, SPSR algorithm, 3D3S modeling, and Geomagic modeling methods were integrated. A refined digital twin model and an updated structural theoretical model for the umbrella-shaped tensile membrane structure are constructed. This enables visualization of spatial geometric form deviations in the entire domain of the umbrella-shaped tensile membrane structure.
3. Based on the accurately reconstructed three-dimensional model of the umbrella-shaped tensile membrane structure, full-life morphological deviation monitoring can be achieved by using Geomagic software to efficiently and accurately obtain spatial geometric form deviation among the physical tensile membrane structure and the original/updated theoretical model.
4. Through finite element nonlinear analysis and comparison with the original/updated theoretical models under load-combination conditions, the differences in full-field membrane stress distribution were analyzed. This method provides a refined model reconstruction and analysis basis for life-cycle health monitoring and inspection of physical tensile membrane structures and supports digital twin-based life-cycle health monitoring and safety inspection of tensile membrane structures.
5. Due to the synergistic influence of various factors in the tensile membrane structure, such as the umbrella-shaped tensile membrane structure case in this paper, the diagonal gusset plates have inconsistent stretching degrees, resulting in significant deviations in the diagonal configuration of the membrane surface. Furthermore, insufficient uplift of the membrane surface leads to the umbrella-shaped tensile membrane structure not reaching the ideal tension state. Therefore, in actual construction projects, it is necessary to attach great importance to the impact of construction errors on the overall spatial form of the membrane surface in tensile membrane structures.
6. This work still has several limitations regarding the generalizability of its findings. Firstly, the investigation was conducted using a single case study, and the experimental validation was limited to a bench-scale specimen. Secondly, only one material was examined. Therefore, the reported performance may not be directly applicable to structures with different dimensions, materials, geometric characteristics, or environmental conditions. Future studies should include larger-scale experiments, multiple material types, and more diverse practical scenarios to further evaluate the robustness and general applicability of the proposed method.

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